

TERM PROJECT PAPER ON FOAM INJECTION MOLDING

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Abstract

Foam injection molding is a specialized process of injection molding that can be a great way to save materials and production costs in the manufacturing industry. This paper examines recent developments in foam injection molding, and research was compiled to discuss its history, process, and applications. Although foam injection molding seems simple on the outside and requires minimal setup, the inability to simulate outcomes can make the process difficult to perfect. Using either physical processes, with additives in the form of supercritical fluid, or chemical processes, with powder additives that give off gas in the mold, foam injection molding can be used across a variety of industries. Current research is working towards refining surface quality and further decreasing part density, but with these issues solved, foam injection molding is sure to expand and become more prevalent in the injection molding and manufacturing world.

Introduction

Injection molding as a whole is an incredibly important and popular form of manufacturing, taking up over 32% of plastics manufacturing. As a relatively simple process, parts can be made extremely quickly by injecting plastic melt into a thin cavity, allowing it to cool, and ejecting the part. Being an incredibly large scale process, there has been lots of research done on optimizing process parameters and increasing part usage. There are also many specialized processes that allow injection molding to span many more industries. A few examples of these are gas-assisted molding, overmolding, and foam injection molding. Foam injection molding has been more prevalent recently, mostly for consumer markets since the early 2000s. Its material-saving qualities promote cheaper production and create parts thicker than the typical injection molding maximum. Since this process is newer, it has not yet been perfected, and there are multitudes of research projects happening that will allow foam injection molding to become a more useful and versatile process with minimal disadvantages.

History and Development

Foam injection molding (FIM) started roughly in the 1950s, when manufacturers added a small pinch of baking soda to the polymer melt to help combat sink marks [1]. This addition allowed the melt to swell slightly in the mold, and the process later became known as chemical foam injection molding. As the use of these additives continued, manufacturers started to notice a lower density in their parts, ranging from 0.3 to 0.6 g/cm³, leading to less material being used and further research to be developed [1].

Researchers were trying to determine how to best control the swell of the material and create smaller and more evenly distributed air bubbles, called cells. In the 1970s, physical propellants were introduced. This process involved gas being injected directly into the polymer melt, as opposed to forming once in the mold cavity. For a while, chemical processes were still favored due to their ease of use, but the physical propellants created more desirable densities.

In the late 1980s, a new process was invented at the Massachusetts Institute of Technology (MIT), called the MuCell process. Originally used for extrusion molding, the MuCell process was a physical foaming process involving gas charging. It was used for injection molding starting in the year 2000, with a supercritical fluid homogenizing with the plastic melt solution and separating in the mold cavity to form a foamy interior. With the MuCell process, researchers were able to achieve a weight reduction as high as 33%, making further research highly desirable [2]. Over the years, this process, along with chemical foaming, was refined and altered, and now it creates parts that are stronger, lighter, and cost-effective.

Process Description

Foam injection molding is defined as a polymer solution that, when injected into a mold cavity, forms a rigid outer skin with a foamy, lightweight interior composed of gas-filled bubbles, or cells. Injection into the mold cavity causes a rapid pressure drop, allowing the gas to disperse while remaining trapped in the part. Figure 1 [3] shows how these cells form and have different disbursements at different cross sections.

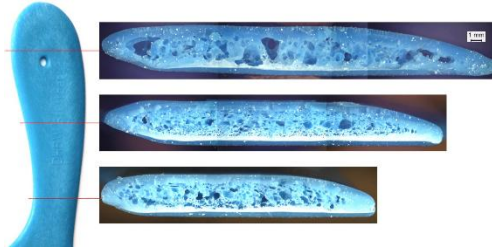


Figure 1: Cell formations in a comb handle

These cells can be made either through physical or chemical processing, the two main types of foam injection molding.

Physical FIM means that a gas additive is directly combined with the polymer melt. Typically nitrogen or carbon dioxide, the high injection pressure allows bubbles to evenly disperse throughout the melt [3]. This process does require a different injection molding machine with a gas jet and a specially designed foaming screw [4]. This screw has a shear section that allows the additive, in the form of a supercritical fluid, to homogenize with the melt before reaching the mold cavity. A non-return valve and a shut-off nozzle are also pertinent to prevent early expansion or gas release in the melt [1]. The pressure drop once reaching the cavity separates the gas from the melt to create the foamy interior, and finely distributed cells that range from 5-50 μ m in diameter [3]. These small bubbles allow for more precise parts, such as electronics and medical equipment.

Chemical FIM relies more on the makeup of the polymer melt solution. While more difficult to control, different add-ins are considered to give off different foam makeups. These chemical blowing agents are added to the melt in a granulate or powder form and are thoroughly mixed with the polymer melt, sometimes changing its thickness or color [5]. After injection into the melt cavity, the heat of the mold itself allows the additive to react and emit gas, creating the foamy interior to the part. The cells made from this chemical reaction are much larger than those of physical foaming, ranging from 50 to 500 μ m, and their distribution is dependent on how the additive was mixed with the polymer melt, as well as the melt front flow into the mold cavity [3]. Since chemical foam injection molding does not require any extra machine setup and can create larger bubbles, it requires less material and is more widely used to make thicker parts with higher structural capabilities [3].

Applicable Materials

The biggest difference between regular and foam injection molding is the additives that go into making the foam. For both chemical and physical foaming, these additives must be non-toxic, non-corrosive, and

non-flammable [6]. This will allow the part to retain its rigidity and not interfere with the polymer itself. Physical FIM requires a supercritical fluid to mix with the polymer melt. This additive must have good thermal stability to reach this state and must not cause any gaseous state chemical reactions. Typical choices include carbon dioxide, nitrogen, or air. Chemical FIM is slightly different, as it requires a chemical reaction to work. Additives such as sodium bicarbonate, ammonium carbonate, or azoformamide all give off gases, including nitrogen and carbon dioxide, when exposed to heat [6]. Since these are added in a powder form, it is imperative that good mixing occurs. Without proper mixing, streak marks or swirls of discoloration on the surface of the parts will appear.

As for the polymer melt itself, foam injection molding originally used standard polymers, such as polyethylene or polypropylene, only, but has expanded to include engineering plastics, high temperature resins, and composites up to 40% glass fiber [3]. Thermoplastics and thermosets can be foam injection molded, but thermoplastics are much more common due to their recyclability and mold machining ease. This base material must be viscous enough to allow the additives to move and foam, while still cooling fast enough to develop a smooth outer skin.

For both chemical and physical injection molding, the additives are relatively cheap and available to purchase. With the addition of gas in the part, FIM polymer savings are typically 5-15% compared to regular injection molding [3]. Some organic agents are also used in chemical FIM, making a good, eco-friendly option. With its wide options for polymers, many foam injection-molded pieces can be recycled, making FIM an inexpensive and accessible process.

Design and Tooling Considerations

Foam injection molding is a relatively easy special process of injection molding, due to the simplicity of implementation. For chemical FIM, there is no difference in tooling from that of a typical injection molding machine. The material, with its chemical additives, is added into the hopper as usual and is mixed thoroughly by the time it reaches the mold cavity. Physical FIM requires a few more changes, including a gas injection system, a plasticizing screw, a non-return valve, and a shut-off nozzle [1]. Figure 2 shows what the setup could look like for physical FIM [7].

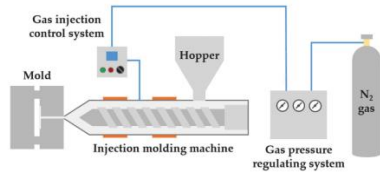


Figure 2: Physical FIM Machine set up

The plasticizing screw has a shear section, allowing for easy and fast mixing with the gas, homogenizing before reaching the mold [4]. The geometry also allows for the second non-return valve to be eliminated from the machine, allowing the plasticizing zone to be slightly longer. The remaining nonreturn valve prevents premature expansion and, paired with the shut-off nozzle, blocks flow from moving backwards in the screw [1].

Other enhancements to the machine and tooling could be considered for specialized processes. Combining foam injection molding with co-injection molding would allow a non-foaming outer shell to be injected into the mold cavity first. Greatly increasing the cost of the machine, this process would require an entire second melt screw and injection unit.

As for the mold itself, foam injection molding is commonly used for larger parts, especially due to the material and weight it saves. Larger parts need larger molds; however, due to the 40% lower clamping force required, the mold and machine can be small, or just large enough to actually mold the part [3].

Cooling channels in foam injection molding are typically part-conforming, meaning they move with the part, as opposed to being straight channels, and can increase the cost of the mold.

The thickness of foam injection molding is also special. Regular injection molding typically has pieces 1 to 4 mm in thickness, whereas chemical FIM can create pieces over 6 mm thick [8]. However, the thickness of the part greatly alters the amount of foaming that can occur. A lower wall thickness will decrease the cooling time in the part. For chemical FIM, this cooling time is what causes the bubbles to form. Therefore, a shorter cooling time would decrease the bubbles formed and decrease the weight reduction in the part, making the process almost the same as basic injection molding. The thickness of these parts allows for better structural capabilities, increasing their versatility.

Application Areas

Foam injection molding spans a wide range of industries, and parts are used for their lightweight and high strength. The automotive sector uses FIM parts for interior door components and dashboards. These pieces are sometimes paired with another form of injection

molding to add the outer skin and texture. Physical FIM, with its small cells, can be used for smaller parts and is popular for smartphone housings and laptop bezels, keeping thin, lightweight walls to house electrical components [3]. Larger-scale products are also made with FIM to help save material. This includes shelving, fencing, storage sheds, and other consumer products [9]. Due to the parts' high temperature stability, the medical industry uses FIM parts for medical device housing and structural components of hospital beds [9]. There are many uses for foam injection molded pieces, and they are commonly used as lightweight structural parts or cheap options for storage components.

Advantages and Disadvantages

Foam injection molding is best known for its ability to save material and create lightweight parts, but overall has economic, technical, and quality advantages and disadvantages.

Starting with economics, foam injection molding saves a lot of material, especially for larger parts. The foamy interior is filled with gas, which is also cheap. A low mold temperature and clamping force are needed, allowing for less energy and a smaller injection molding machine to be used. Moreover, chemical foam injection molding requires no extra mold or machine tools, causing lower startup costs. On the other hand, physical foam injection molding does need special machinery, which can be more expensive for short-term runs, but would be beneficial on a long term scale.

The parts themselves are relatively easy to make, and can have higher wall thicknesses than normal injection-molded parts. This increases the use of FIM parts and expands their industrial abilities. With physical FIM having small, dispersed cells and chemical FIM giving larger cells, parts can be made of many different sizes and geometries. One of the greatest difficulties of the process, however, lies in the process parameters. Since physical FIM injects gas directly into the melt, these parameters are relatively easy to change. Chemical FIM is much different, and the size and spread of the bubbles rely on the melt mixture and additive formulation, which is much more difficult to make quick changes to. Either way, foam injection molding is not easily simulated, meaning most processes run on a trial-and-error dependence, creating a long process to fine-tune parts.

Lastly, the part quality itself must be discussed. Foam injection molded parts are desirable for their high strength-to-weight ratio, with little warpage. However, adding air into these parts decreases the impact strength, and the parts are susceptible to cracks [10]. Surface quality is also a big issue if bubbles form on the surface before it develops an outer skin. For Chemical

FIM, surface defects also include swirl marks or discoloration, typically due to uneven mixing of the additive. That being said, there is a lot of current research to improve the surface quality of foam injection molded parts, which will allow their market to grow even more over the coming years.

Future Developments

Multiple research projects are currently being run on foam injection molded parts, the most prominent ones being to improve the surface quality and further decrease the size of the cells.

Even though foam injection molding started as a way to avoid sink marks, it caused other issues, such as bubble marks from escaped gas and swirl marks from improper mixing. Figure 3 shows some of these surface defects, including silver marks from gloss variation, and swirl marks.

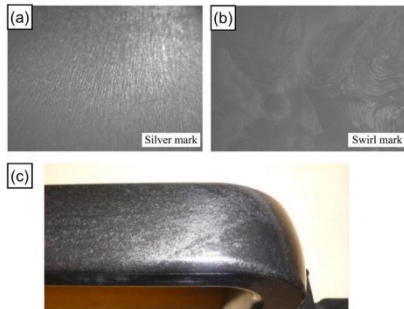


Figure 3: Surface Defects on FIM Parts

Due to rapid solidification on the cavity walls, there is little time to avoid defects, which can cause many injection issues [11]. There are specific calculations and adjustments to help combat this, but this can take a long time to perfect without efficient simulation tools, so some other, more costly methods have been tested.

One of these methods is gas counterpressure. Applying high air and gas pressure in the sealed cavity combats the bubble cell formation and can improve the surface gloss dramatically [11]. This means that surface bubble formation can be minimized, and the gas is trapped in the melt until a solid outer skin forms. The gas is then released to give the foamy interior without compromising surface quality. Issues arise in this process as it is much more complex than basic foam molding. Also, depending on how the part cools before the pressure drop, the interior foam could be unevenly dispersed around the cavity. In molding without counterpressure, the melt front tries to avoid any resistance and first fills the nominal walls. The thinner walls and excess features are filled at the end of the sequence. Counterpressure, shown in Figure 4, controls the resistance on the flow front and causes the melt to

enter the thinner features first, before the foaming takes place [12].

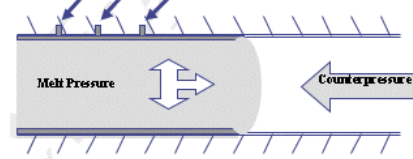


Figure 4: Counterpressure against the melt front

Another method to help improve surface quality is the use of co-injection molding. This process has a plain polymer melt injected into the mold cavity first to form an outer layer. The foaming material is then injected, keeping it in the center of the part. Similar to gas-counterpressure, there could be issues with the location and evenness of the foam, but this process is more likely to give a better surface layer. The option to have multiple colors or more eco-friendly materials also arises, but this process would require much more material than plain foam injection molding.

Another current study is to create a foam composed of nano cells. Smaller cells allow the part to have a higher strength-to-weight ratio, as well as increased thermal insulation. Physical foam injection molding can create cells as small as $5\mu\text{m}$, but reducing this to $1\mu\text{m}$ would allow parts to have even lower weight and density.

A research project from the ASME Journal of Micro and Nano Science and Engineering was studying the effects of electromagnetic foaming injection forming [13]. The article recognized that nanocellular foaming was only apparent in batch processing, but would be beneficial to bring to industrial production facilities. The experiment consisted of a thin metal sheet being inserted in the mold, which the melt will push to be set with the mold. Shown in Figure 5, “Right after the injection, a high-intensity electromagnetic field is applied inside the cavity, causing additional deformation of the sheet metal. This additional deformation not only finalizes the geometry but also causes a sudden pressure drop inside the cavity, which initiates the foaming process in the plastic melt” [13].

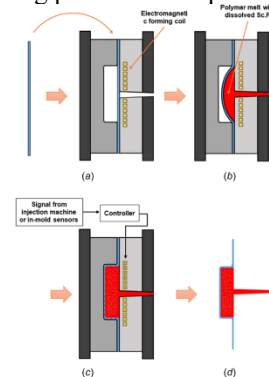


Figure 5: Electromagnetic foaming process

Many tests were run, and in the end, it was determined that the main impact on the size of the bubbles was the strength and duration of the pressure drop. The steeper the drop, the smaller and more uniform the cells formed. The study determined that this process would work on a larger scale, but would cost a lot and be difficult to implement. However, if a lightweight and durable material were used, this process would be beneficial for the injection molding industry [13].

As both of these research areas are newer, it would take a bit longer to fully adopt in industry. They both require much higher start-up costs, although with the general low cost of foam injection molding, it may not be much of an issue for high-demand parts. Either way, this research is sure to expand foam injection molding in the future.

Conclusion

Overall, foam injection molding is a great process that can save material, money, and time to effectively produce parts. It spans many different types of materials and additives, each creating different strength profiles that allow for a wide range of parts, from medical to structural. With minimal machine changes, parts are easy to set up, which is helpful considering its difficulty of modeling before physical testing. Surface quality is one of the biggest issues faced with FIM, but more research is coming out to help improve this, or pair FIM with co-injection molding to have a fully separate outer layer. Altogether, originating from simply fixing sink marks, foam injection molding is now a complex practice that makes many parts used every day.

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